Essentials of Probability CheatSheet

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Introduction

What is Probability?

Definition: Probability is the branch of mathematics that measures how likely an event is to happen. It quantifies uncertainty and helps us make predictions about various outcomes. The probability of an event is expressed as a number between **0 (impossible)** and **1 (certain).**

Purpose: Probability helps answer questions like:

- How likely is it to rain tomorrow?
- What are the chances of getting stuck at a traffic light?
- What is the probability of a machine learning model making a mistake?

Randomness and Experiments:

- **Randomness**: Refers to situations where we can't predict outcomes in advance.
  - Example: Flipping a coin.
- **Experiment**: A procedure involving random events that leads to an outcome.
  - Example: Tossing a coin 10 times and observing the outcomes.

Key Terms:

- **Trial**: Each instance in an experiment (e.g., each coin flip).
- **Outcome**: The result of a trial (e.g., heads or tails).
- **Experiment**: A series of trials that produce multiple outcomes.

Coin Toss Example:

- If you toss a fair coin 10 times:
  - Possible outcomes are Heads **(H)** or Tails **(T)**.
  - A real experiment might result in HTTTTTTHHT (7 tails, 3 heads).
  - A random-looking guess might be THHTTTHTHH (5 heads, 5 tails).

Applications in Machine Learning:

- Probability is used to deal with uncertainty in data, helping in prediction models, error rates, and decision-making processes.

Key Probability Concepts

Order and Structure in Probability

- **Hidden Patterns**: In probability, we often search for hidden structures and patterns in seemingly random events. This concept is crucial in fields like machine learning, where understanding the underlying probabilities can reveal trends in data.

- **Randomness**: When we refer to an event as random, it means that its outcome cannot be predicted with certainty, but specific patterns or structures may still exist. Without these structures, probability and statistics wouldn't work effectively.

Example 1: Dice Game in a Casino

- **Scenario**: You are playing a dice game where you guess whether the result of a six-sided die roll will be even or odd.
  - The numbers on the die are **{1, 2, 3, 4, 5, 6}**.
  - Evens: {2, 4, 6},
  - Odds: {1, 3, 5}.
  - **Number of even outcomes = 3**.
  - **Total possible outcomes = 6**.

- **Probability of Winning**:
  - The probability of guessing correctly (even or odd) is $^{3}/_{6} = ½$

Example 2: Picking a Random Pen

- **Scenario**: You randomly pick a pen from a drawer with 5 blue pens, 3 red pens, and 2 black pens.
  - **Total pens = 5 (blue) + 3 (red) + 2 (black) = 10 pens.**

- **Probability of Picking a Black Pen**:
  - Number of black pens = 2.
  - Total possible outcomes = 10.
  - **Probability = $^{2}/_{10}$ = 0.2 or 20%.**

Randomness and Patterns

- **Perceived Randomness**: Our intuition might think certain patterns (like HTHTHTHT) are more random than others (like HHHHHHHH). However, both sequences have an equal chance of occurring in a fair coin flip experiment.

- **Real vs. Perceived Randomness**:
  - **Actual flips**: Example: HTTTTTTHHT (7 tails, 3 heads)
  - **Random-looking guess**: Example: THHTTTHTHH **(5 heads, 5 tails)**
  - Both outcomes are equally random, even though the first sequence may not *feel* random.

Randomness in Coin Flips

Definition: Randomness refers to the unpredictability of individual outcomes in an event. In the context of coin flips, each flip is independent, and the result **(heads or tails)** cannot be predicted in advance.

Key Concepts:

1. **Fair Coin**: A coin is considered "fair" if both outcomes (heads and tails) are equally likely. The probability of each outcome is:
    - P(Heads) = **0.5**
    - P(Tails) = **0.5**

2. **Independent Events**: In coin flips, each flip is independent of the previous ones. This means the outcome of one flip doesn't affect the outcome of the next.
    - **Example:** If you flip heads **5 times** in a row, the next flip still has a **50%** chance of being heads.

3. **Experiment**: A coin flip experiment involves repeating the action **(flipping a coin)** multiple times to observe randomness over several trials.

- **Example:** Flipping a coin 10 times might give a result like HTTHHHTTTT. This sequence will vary every time you repeat the experiment.

Why Randomness Matters in Probability:

- Without randomness, predicting outcomes would be too easy, and probability theory would lose its purpose.
- Randomness ensures that while individual outcomes are unpredictable, the overall pattern across many flips **(like the probability of heads being 50%)** follows predictable rules.

The Sample Space and Events

- **Sample Space (S)**: The set of all possible outcomes in a probability experiment.
  - **Example 1:** Rolling a six-sided die, the sample space is **S={1,2,3,4,5,6}**.
  - **Example 2:** Flipping a coin, the sample space is **S={H,T}**.
  - **Example 3:** Waiting for a pot to boil, the sample space is **S=[0,∞)**, meaning any amount of waiting time is possible.

- **Event (E)**: A subset of the sample space that represents a specific outcome or a group of outcomes.
  - **Example 1:** Rolling a die, an event might be **E={2,4,6}** (rolling an even number).
  - **Example 2:** Flipping a coin, possible events are **E={H},{T},{H,T},{}** (getting heads, tails, both, or neither).
  - **Example 3:** Waiting time event might be E=**{between 2 and 5 minutes}**.
- **Probability of an Event**: The likelihood of an event **E** occurring is written as **P(E)**.
  - **Example 1:** The probability of rolling a 1 on a die is **P({1})=$^1/_6$**
  - **Example 2:** The probability of rolling an even number **E={2,4,6}** is **P(E)**= **$^3/_6$ =$^1/_2$**.

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Sample Space of Three Coin Flips:

- When flipping a coin **three times**, the sample space would include all possible combinations of heads and tails:
  - **S= {HHH , HHT , HTH , HTT , THH , THT , TTH , TTT}**

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Applications of Probability

Machine Learning Use Cases

- **Exploratory Data Analysis**: Sampling data to understand patterns and distributions.

- **Bias/Variance Tradeoff**: Used in model evaluation to balance underfitting (bias) and overfitting (variance).

- **Prediction**: Forecasting unknown values based on observed data relies heavily on probability.

Statistics and Probability:

- **Statistics** helps interpret the results of testing machine learning models and make better decisions in business and finance.

- **Probability** provides the foundation for understanding statistics and its tools.

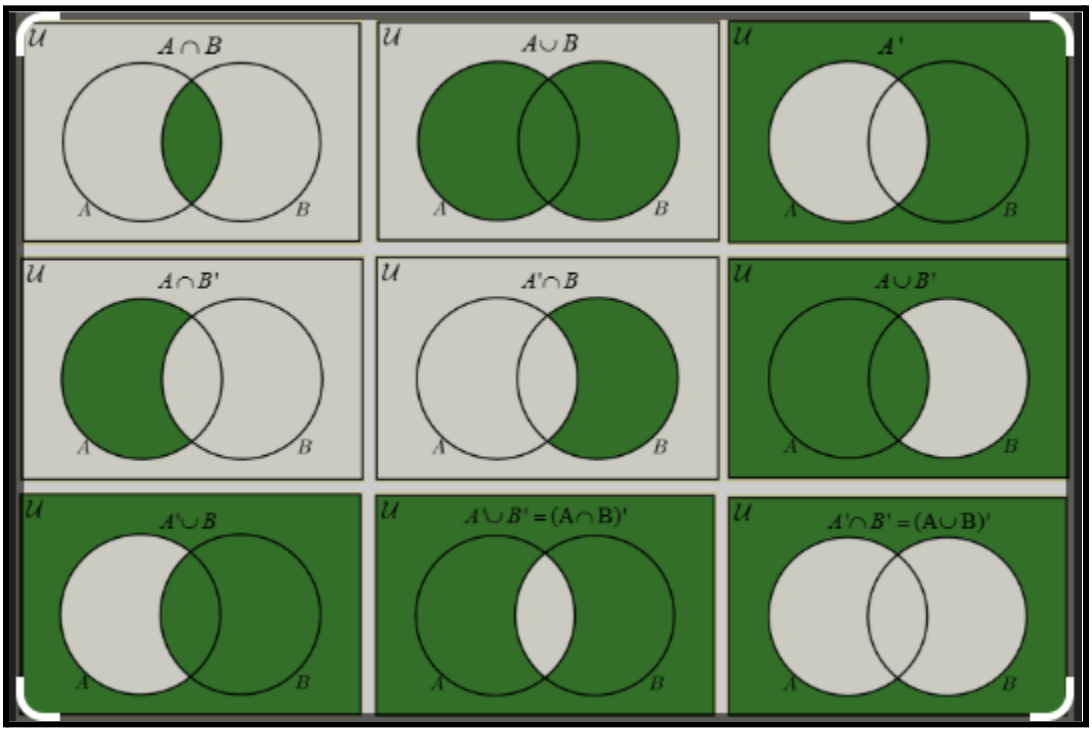
Real-Life Example: Spam Mail Probability:

- **Problem:** You receive a lot of mail each day, and the spam rate is **39.3%.** What's the probability that a randomly selected mail is not spam?
    - **Solution:**

- Probability of not spam $P(\text{Not Spam}) = 1 - P(\text{Spam})$.
- $P(\text{Not Spam}) = 1 - 0.393 = 0.607$, or **60.7%** chance that the mail is not spam.

Axioms of Probability

The **axioms of probability**, introduced by Andrey Kolmogorov in 1933, are the rules that govern all probability theory. These form the foundation for calculating and understanding probabilities.



Axiom 1: Non-Negativity

The probability of any event E is always between **0 and 1 (inclusive)**. This means that no event can have a negative probability, and an event with a probability of **1 is certain to occur.**

$$0 \leq P(E) \leq 1$$

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Example: The probability of flipping heads in a coin toss is **P(Heads)=0.5** This falls within the allowed range of **0 to 1.**

Axiom 2: Certainty of the Sample Space

The probability of the entire sample space **S**, which includes all possible outcomes, is **always 1**. This means something in the sample space is guaranteed to happen. **P(S)=1**

Example: In rolling a six-sided die, the probability of getting any result **(1, 2, 3, 4, 5, or 6)** is **1** because one of these outcomes must occur.

Axiom 3: Additivity of Mutually Exclusive Events

If two events **A** and **B** are mutually exclusive (meaning they cannot occur at the same time), the probability of either event occurring is the sum of their individual probabilities. **$P(A \cup B) = P(A) + P(B)$**

Example: In a dice roll, the probability of getting either a 2 or a 4 (mutually exclusive events) is

$P(\{2\}) + P(\{4\}) = {}^{1}/_{6} + {}^{1}/_{6} = {}^{1}/_{3}$

Complementary Events

For any event **E**, the complementary event not **E** is the event that **E** does not occur. The probability of the complementary event is:

$$P(\text{not } E) = 1 - P(E)$$

Example: If the probability of raining tomorrow is 0.3, then the probability that it does **not** rain is: **1−0.3=0.71**

Continuous Probability

Continuous probability deals with events that have an infinite number of possible outcomes, like measuring time, height, or temperature.

Definition: In continuous probability, the outcomes are not distinct (like heads or tails) but can take any value in a given range. We don't calculate probabilities for specific values, but instead for ranges of values.

Example: Waiting time for a bus. If you're waiting for a bus that can arrive at any time between 0 and 20 minutes, the probability of waiting exactly 10 minutes is 0 (because in continuous probability, the likelihood of hitting any exact point is zero). However, you can calculate the probability of waiting between 10 and 15 minutes.

Key Concept: The **Probability Density Function (PDF)** is used to describe the likelihood of different outcomes in continuous probability. The **area under the curve** of a PDF within a range gives the probability for that range.

Two-Dimensional Continuous Probability Example:

- **Scenario**: You and your colleague both want to take a 10-minute coffee break in the next hour. What's the probability that you run into each other at the coffee machine?
    - Both of you have a 60-minute window to take your break.
    - For you to run into your colleague, your break times must overlap within that 60-minute window.
    - After calculating possible overlaps, the probability turns out to be around $^{1}/_{3}$, meaning **you have a 1 in 3 chance of meeting them during your break**.

Estimating π Using Probability

You can estimate the value of π using the **Monte Carlo method** and probability theory. Here's how:

- Imagine a square with a circle inscribed inside it.
  - The square has an area of $\mathbf{4r^2}$ and the circle has an area of $\boldsymbol{\pi r^2}$, where **r** is the radius of the circle.
- Randomly place points inside the square.
  - Some of these points will land inside the circle, and some will land outside but still inside the square.
- The ratio of the number of points that fall inside the circle to the total number of points placed inside the square is approximately $\boldsymbol{\pi/_4}$.
- From this, you can estimate π by multiplying this ratio by 4

$$\pi \approx 4 \times \frac{\text{points inside the circle}}{\text{total points}}$$

- The more points you randomly place, the closer your estimate of π will be.

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Counting Techniques

Counting in Probability

Counting plays a key role in determining the size of the sample space and calculating probabilities. By knowing how many ways an event can happen, we can better understand its likelihood.

Common Counting Techniques

1. **Counting with Replacement (Exponents)**:
    - If repetitions are allowed, we use exponents to count the number of possible outcomes.
    - **Example**: If you are setting a 4-digit ATM PIN, each digit can be any number between 0 and 9, so there are: **$10 \times 10 \times 10 \times 10 = 10^4 = 10{,}000$ possible PINs.**

2. **Counting without Replacement (Factorials)**:
    - If repetitions are **not** allowed, the number of available options decreases with each selection, and we use factorials.
    - **Example**: For a padlock where you choose a 3-number combo from 40 numbers without repeats, the number of combinations is: **$40 \times 39 \times 38 = 59{,}280$**

Factorials (n!):

- A **factorial** is the product of all integers from 1 to n.
  - **Notation**: **$n! = n \times (n-1) \times (n-2) \times \ldots \times 2 \times 1$**
  - **Example**: The number of ways to arrange a **deck of 52 cards is**: **$52! = 52 \times 51 \times 50 \times \cdots \times 1$**

Permutations and Combinations

Permutations:

- **Permutations** are used when the order of outcomes matters.
  - **Formula** for arranging **n** objects: **n!**
  - **Example**: If you want to know how many ways you can arrange **7 letters, with no repeats**, the answer is **$7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5{,}040$**

Combination Lock Example:

- You have a padlock with a **3-number combination from numbers 0 to 39 (no repeats allowed)**. The number of possible combinations is: **40 × 39 × 38 = 59,280**
  This uses the formula for permutations because the order of the numbers matters.

License Plate Example:

- A New York license plate has three letters followed by four digits. What is the probability that both the three-letter string and the four-digit number are palindromes?
  - **Total possible plates**: There are $\mathbf{26^3 \times 10^2}$ possible license plates.
  - **Palindromes**: A palindrome reads the same forwards and backwards. The number of palindrome plates is: $\mathbf{26^2 \times 10^2}$

$$26^2 \times 10^2.$$

Probability:

$$\frac{26^2 \times 10^2}{26^3 \times 10^4} = \frac{1}{2600}.$$

So, the probability of randomly selecting a palindrome plate is $\frac{1}{2600}$.

Permutations with Examples:

- Permutations are useful when the **order** of the selection matters.
  - **Example 1**: Arranging 52 cards in a deck. The number of ways to arrange them is **52!**
  - **Example 2**: Arranging 3 objects out of 5 in a specific order. The number of ways is: **P(5,3)= 5 × 4 × 3 = 60.**

Binomial Coefficients (Combinations)

- **Definition**: A binomial coefficient represents the number of ways to choose k objects from a set of n objects, **ignoring the order**.

- It is denoted as $\binom{n}{k}$, read as "n choose k."
- **Formula**:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Where $n!$ (n factorial) represents the product of all integers from 1 to n, and $k!$ and $(n-k)!$ are factorials as well.

Examples of Binomial Coefficients:

1. **Selecting Students**:
   - **Problem**: How many ways can you choose 5 students from a class of 30 students?
   - **Solution**:

$$\binom{30}{5} = \frac{30!}{5!(30-5)!} = \frac{30 \times 29 \times 28 \times 27 \times 26}{5 \times 4 \times 3 \times 2 \times 1} = 142,506 \text{ ways.}$$

2. **Grid Paths**:
   - **Problem**: In a city with a rectangular grid of streets, how many ways can you go 4 blocks east and 6 blocks north, moving only east or north?
   - **Solution**: This can be solved using binomial coefficients, where you are choosing 4 eastward movements out of 10 total movements (4 east + 6 north):

$$\binom{10}{4} = \frac{10!}{4!6!} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210 \text{ ways.}$$

Inclusion-Exclusion Principle

The **Inclusion-Exclusion Principle** helps calculate the probability of the union of overlapping events by subtracting the over-counted intersections.

1. **Real-Life Loan Example**:
    - **Problem**: You work at an NBFC offering four loans (home loan, car loan, gold loan, education loan). A person can only get one loan, and the probabilities are:
        - P(Home Loan) = **0.2**
        - P(Car Loan)=**0.33**
        - P(Gold Loan)=**0.22**
        - P(Education Loan)=**0.17**
    - **Solution**: Since the events are mutually exclusive (only one loan can be given), the total probability is simply the sum:
    - **P(any loan) = 0.2 + 0.33 + 0.22 + 0.17 = 0.92.**

Monty Hall Problem

- **Scenario**: You are on a game show with 3 doors. Behind one door is a prize, and behind the other two, nothing. You pick a door, but before you open it, the host (Monty) opens one of the other two doors, revealing no prize. He then asks, **"Do you want to switch doors?"**

- **Key Insight**: The probability of your initial choice being correct is $^{1}/_{3}$, and the probability of the prize being behind one of the other two doors is $^{2}/_{3}$. When Monty opens one of the two remaining doors, the probability of the prize being behind the other unopened door increases to $^{2}/_{3}$.
    - **Conclusion**: You should always **switch doors**, as it gives you a higher chance of winning ($^{2}/_{3}$ versus $^{1}/_{3}$).

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Advanced Probability Concepts

Independence and Mutual Exclusivity

- **Independent Events**: Two events **A** and **B** are independent if the occurrence of one does not affect the probability of the other.

  - **Formula**: **$P(A \cap B) = P(A) \times P(B)$**
  - **Example**: Flipping a fair coin twice and getting heads both times: **$P(HH) = {}^{1}/_{2} \times {}^{1}/_{2} = {}^{1}/_{4}$**

- **Mutually Exclusive Events**: Two events **A** and **B** are mutually exclusive if they cannot happen at the same time.
  - **Formula**: **$P(A \cap B)=0$**

    Example: Rolling a fair die:
    - Let **A** be rolling a 6, and **B** be rolling a **2** or **3**.
    - Since **A** and **B** cannot happen simultaneously: **$P(A \cap B)=0$**, **${}^{1}/_{6} + {}^{1}/_{3}=0.5$**

- **Are they Independent or Mutually Exclusive?**
  - **Example**: Rolling a die:

- Let **A** = rolling a 6, and **B** = rolling a 2 or 3.
- **A** and **B** are mutually exclusive because they cannot occur together.
- **P(A∩B)=0** , but they are not independent.

Conditional Probability

- **Definition**: The probability of an event occurring given that another event has already occurred.

- **Formula**:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

- **Example**: You toss a fair coin three times, and you know there's at least one heads. What's the probability of getting at least two heads?
  - Let A = at least two heads, and B = at least one heads.
  - $P(A \cap B) = P(\text{at least two heads}) = \frac{4}{8} = \frac{1}{2}.$
  - $P(B) = P(\text{at least one head}) = \frac{7}{8}.$
  - Conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{2}}{\frac{7}{8}} = \frac{4}{7}.$$

Bayesian Probability

Bayesian probability provides a way to update the probability of an event as new evidence or information becomes available. It's useful when dealing with uncertainty and revising beliefs in light of new data.

1. Bayes' Theorem

- **Definition**: Bayes' Theorem describes how to update the probability of a hypothesis, given new evidence.

- **Formula**:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where:

- $P(A|B)$ = Posterior probability (probability of A, given B)
- $P(B|A)$ = Likelihood (probability of B, given A)
- $P(A)$ = Prior probability (initial belief in A)
- $P(B)$ = Evidence or marginal likelihood (total probability of observing B)

Prior, Likelihood, and Posterior Probability

- **Prior Probability (P(A))**: This is the initial probability or belief about an event before any new evidence is introduced.
  - **Example**: Suppose 5% of the population has a certain disease. Before testing, your prior belief about someone having the disease is 5%.

- **Likelihood (P(B|A))**: This is the probability of observing the evidence, assuming that the hypothesis is true.
  - **Example**: A medical test correctly identifies the disease 99% of the time (i.e., if someone has the disease, the test will be positive 99% of the time).

- **Posterior Probability (P(A|B))**: This is the updated probability of the hypothesis after taking the new evidence into account.
  - **Example**: If the test is positive, you use Bayes' Theorem to calculate the updated probability that the person actually has the disease, considering the prior and the likelihood of the test result.

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Real-Life Cases

Case 1: Covid Testing

- **Problem**: Suppose 1 in 200 people in a population has Covid-19. There is a test that is 99% accurate when someone has the disease (true positive rate) and gives false positives 2% of the time.
- **Solution Using Bayes' Theorem**:
  - **Prior**: $P(\text{Covid}) = \frac{1}{200} = 0.005$.
  - **Likelihood (True Positive)**: $P(\text{Positive Test} \mid \text{Covid}) = 0.99$.
  - **False Positive Rate**: $P(\text{Positive Test} \mid \text{No Covid}) = 0.02$.
  - **Total Probability of a Positive Test**:

$$P(\text{Positive}) = P(\text{Positive} \mid \text{Covid}) \times P(\text{Covid}) + P(\text{Positive} \mid \text{No Covid}) \times P(\text{No Covid})$$

$$P(\text{Positive}) = (0.99 \times 0.005) + (0.02 \times 0.995) = 0.02485$$

  - **Posterior Probability**:

$$P(\text{Covid} \mid \text{Positive Test}) = \frac{0.99 \times 0.005}{0.02485} \approx 0.1991 \text{ or } 19.91\%.$$

  - **Interpretation**: Even with a positive test result, there's only about a 20% chance the person actually has Covid, due to the rarity of the disease in the population.

Case 2: Defective Bolts in a Factory

- **Problem**: In a factory, there are three machines M_1, M_2, and M_3 that manufacture 2000, 2500, and 4000 bolts per day, respectively. The percentage of defective bolts from each machine is 3%, 4%, and 2.5%, respectively. A randomly selected bolt is found to be defective. What's the probability that it was produced by machine M_2?
- **Solution Using Bayes' Theorem**:
  - **Prior**:

$$P(M_2) = \frac{2500}{2000 + 2500 + 4000} = 0.3125.$$

  - **Likelihood**: $P(\text{Defective} \mid M_2) = 0.04$.
  - **Total Probability of Defective Bolt**:

$$P(\text{Defective}) = P(\text{Defective} \mid M_1)P(M_1) + P(\text{Defective} \mid M_2)P(M_2) + P(\text{Defective} \mid M_3)P(M_3)$$

$$P(\text{Defective}) = 0.03 \times \frac{2000}{8500} + 0.04 \times \frac{2500}{8500} + 0.025 \times \frac{4000}{8500} = 0.03176.$$

  - **Posterior Probability**:

$$P(M_2|\text{Defective}) = \frac{0.04 \times 0.3125}{0.03176} \approx 0.393.$$

  - **Interpretation**: Given that the bolt is defective, there is a 39.3% chance it came from machine M_2.

Conclusion

Throughout this document, we have explored the foundations of probability and its real-life applications.

Probability serves as a crucial tool in dealing with uncertainty, whether in daily life scenarios or in advanced fields like machine learning and data science. Understanding core concepts like sample space, events, independent and mutually exclusive events, conditional probability, and Bayesian probability enables us to better interpret outcomes and make more informed predictions.

With examples ranging from coin flips to complex Bayesian probability, we've seen how randomness and structure work together in real-world applications. By applying these concepts, especially in fields like machine learning, you can enhance your predictive models and decision-making processes. Ultimately, probability theory provides the framework to tackle uncertainty systematically, from business decisions to everyday occurrences.

Resources

Books:

- *Probability and Statistics for Engineers and Scientists* by Ronald E. Walpole
- *An Introduction to Probability Theory and Its Applications* by William Feller
- *Essentials of Probability* by Eric Brattain

Research Papers:

- *Bayesian Networks in Machine Learning* by David Heckerman
- *Probability, Statistics, and Machine Learning* by Kevin P. Murphy

Additional Resources:

- Monty Hall Problem Explained
- Monte Carlo Method
- Inclusion-Exclusion Principle